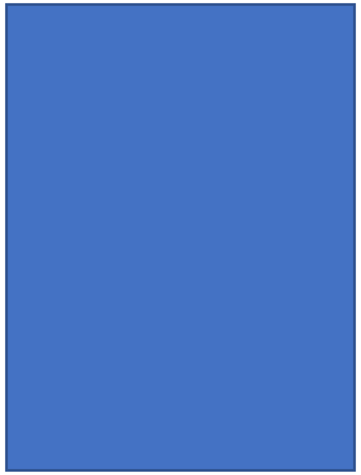
# Feature selection

# When do we need feature selection and why?

In a large dataset, #features can be large:



- Higher computational cost (training time, energy usage)
- Difficult to generalise
- Difficult to see important feature
- Less interpretability

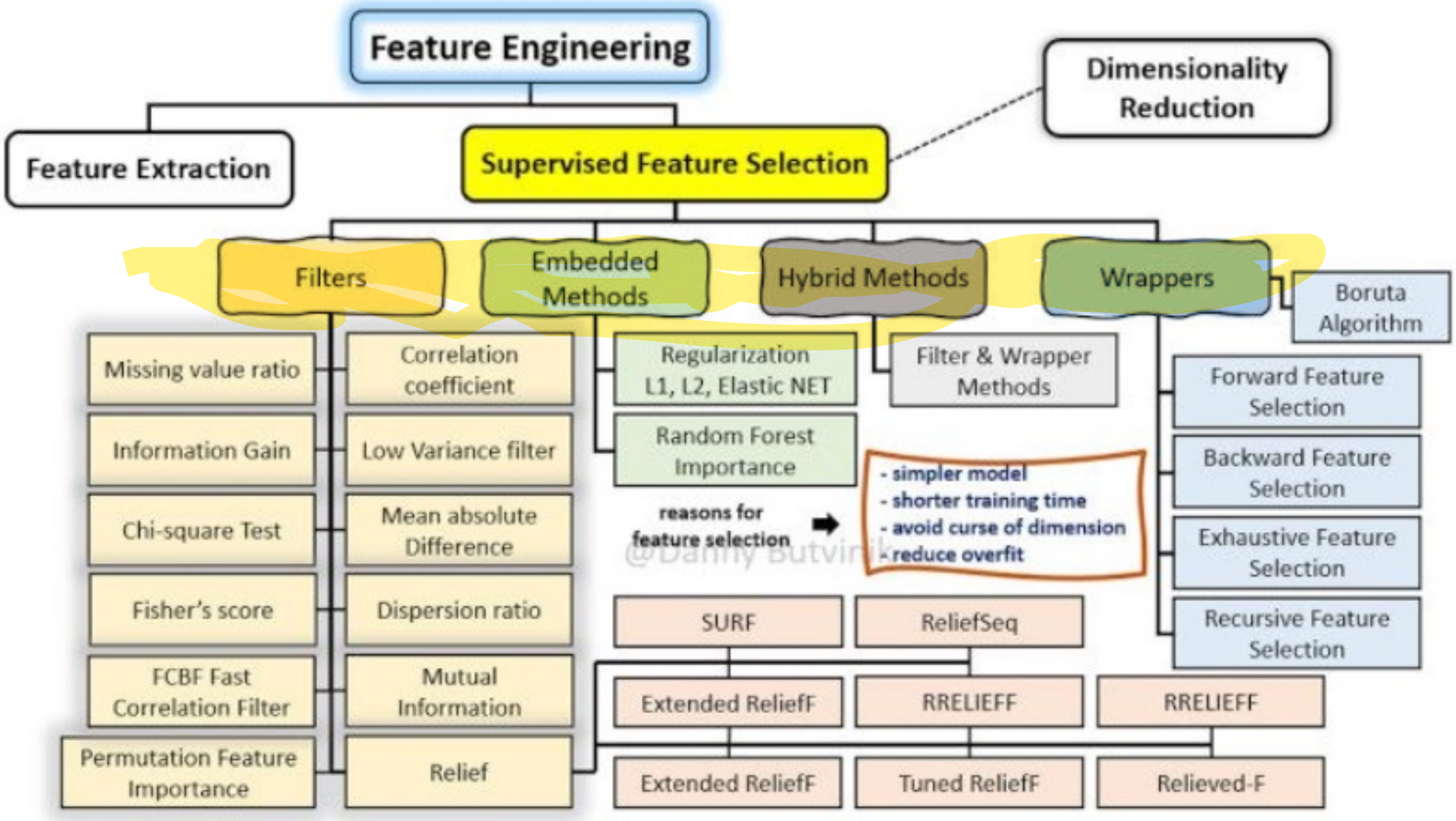
In dataset, where #features is larger compared to #samples

- Overfitting

# Goals of feature selection

- Select most relevant and non-redundant features
- Reduce computational cost
- Simplify model, increase interpretability
- Reduce overfitting, increase generalisation
- Avoid curse of dimensionality

# What methods can be used for feature selection for supervised learning?




[Source](#)

# Filter methods

- How?
  - use statistical measures to rank the features,
  - then SelectKBest, SelectPercentile
- - each feature is considered in isolation, interaction between features is not considered
  - Will cut features which are useless individually but useful when combined with other feature(s)
  - Does not know if two features are identical
- - independent of ML alg.
- +less computational time
- +avoid overfitting

# Wrapper methods

- Essentially: a search strategy (best first search) + search space operator (forward to add feature, backward to delete feature, or bi-directional) + an ML alg.
- -performance depends on the chosen ML alg.
- -greedy alg., may not find optimal subset
- -can still be computationally expensive
- +take into account interaction between features
- include ML to select the feature
- +no redundant features included

- Exhaustive feature selection

Intractable to do! *n* features leads to 2^*n* possible sets of features to train and evaluate.

**for** every possible set of features $S$:
  train and evaluate the model based on $S$
**return** the set $S_{max}$ that gave the best result

- Forward feature selection (SFS)

$S$ = empty set
**repeat**:
  find the feature $F$ that gives the largest improvement when added to $S$
  if there was an improvement add $F$ to $S$
**until** there was no improvement

- Backward feature selection

Same like SFS, but start from full set and then remove

- Recursive feature selection

# Embedded methods

- Feature selection is part of the training process of the ML alg.
- Examples:
  - Feature importance in Random forest or Gradient Boosting
  - Regularisation for linear modela (e.g., $L_1$ Lasso)
- +take into account interaction between features
- +faster than wrappers
- +more accurate like filters

# Hybrid methods

- Hybrid of filter and wrapper methods

- How?
    - use ranking methods to get a feature ranking list – to reduce feature space
    - then apply wrappers on top k features from the ranking list

# See notebook!